8.4 Trigonometric Substitutions

- For integrals involving
  $\sqrt{a^2 - x^2}$, $\sqrt{a^2 + x^2}$ and $\sqrt{x^2 - a^2}$, where $a > 0$.

- Three basic trigonometric substitutions are
  $x = a\sin\theta$, $x = a\tan\theta$ and $x = a\sec\theta$. to eliminate $\sqrt{\ }$

When a substitution is done, we need it to be reversible, and by recalling inverses of trigonometric functions, a substitution can be done under certain domain constraints.

$\theta = \sin^{-1}\left(\frac{x}{a}\right)$

- Using the substitution of $x = a\sin\theta$ in $\sqrt{a^2 - x^2}$, we require $-\frac{\pi}{2} \le \theta \le \frac{\pi}{2}$.

$dx = a\cos\theta\, d\theta$

$\sqrt{a^2 - x^2} = \sqrt{a^2 - a^2\sin^2\theta} = \sqrt{a^2(1 - \sin^2\theta)} = a\sqrt{\cos^2\theta} = a|\cos\theta| = a\cos\theta$
$(\because \cos\theta \ge 0)$

- Using the substitution of $x = a\tan\theta$ in $\sqrt{a^2 + x^2}$, we require $-\frac{\pi}{2} < \theta < \frac{\pi}{2}$. $\theta \in [-\frac{\pi}{2}, \frac{\pi}{2}]$

$dx = a\sec^2\theta\, d\theta$

$\sqrt{a^2 + x^2} = \sqrt{a^2 + a^2\tan^2\theta} = \sqrt{a^2\sec^2\theta} = a|\sec\theta| = a\sec\theta$
$(\because \sec\theta > 0,\ \text{when } -\frac{\pi}{2} < \theta < \frac{\pi}{2})$

- Using the substitution of $x = a\sec\theta$ in $\sqrt{x^2 - a^2}$, we require $0 \leq \theta < \frac{\pi}{2}$ if $\frac{x}{a} \geq 1$ and $\frac{\pi}{2} < \theta \leq \pi$ if $\frac{x}{a} \leq -1$. (For simplicity, we assume that $0 \leq \theta < \frac{\pi}{2}$, $\frac{x}{a} \geq 1$)

$dx = a\sec\theta\tan\theta\, d\theta$

$$\sqrt{x^2 - a^2} = \sqrt{a^2\sec^2\theta - a^2}$$
$$= \sqrt{a^2(\sec^2\theta - 1)}$$
$$= a|\tan\theta| = a\tan\theta$$

we assumed $\theta \in [0, \frac{\pi}{2})$, where $\tan\theta \geq 0$.

Remark:

- For $\sqrt{a^2 - (bx)^2}$, substitute $x = \frac{a}{b}\sin\theta$.
- For $\sqrt{a^2 + (bx)^2}$, substitute $x = \frac{a}{b}\tan\theta$.
- For $\sqrt{(bx)^2 - a^2}$, substitute $x = \frac{a}{b}\sec\theta$.

$\sqrt{35 - 3x^2}$

$35\left(1 - \frac{3}{35}x^2\right)$

$\frac{3}{35}x^2 = \sin^2\theta$

$x^2 = \frac{35}{3}\sin^2\theta$

$\Rightarrow x = \sqrt{\frac{35}{3}}\sin\theta$.

Example

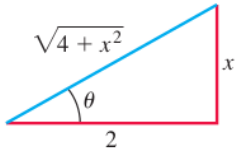
$$\int \frac{dx}{\sqrt{4+x^2}} = \int \frac{2\sec^2\theta\, d\theta}{2\sec\theta} = \int \sec\theta\, d\theta \to \int \sec\theta \frac{\sec\theta+\tan\theta}{\sec\theta+\tan\theta} d\theta$$

$x = 4\tan^2\theta$

$\Rightarrow x = 2\tan\theta \Rightarrow \tan\theta = \frac{x}{2}$

$dx = 2\sec^2\theta\, d\theta$

$\sqrt{4+x^2} = \sqrt{4(1+\tan^2\theta)} = 2\sec\theta$



$\Rightarrow \sec\theta = \frac{\sqrt{4+x^2}}{2}$

$= \ln|\sec\theta+\tan\theta| + C$

$= \ln\left|\frac{\sqrt{4+x^2}}{2} + \frac{x}{2}\right| + C$

$= \ln|\sqrt{4+x^2}+x^2| + D$

$\ln|\sqrt{4+x^2}+x| - \ln 2 + C$

FIGURE 8.4 Reference triangle for $x = 2\tan\theta$ (Example 1):

$$\tan\theta = \frac{x}{2}$$

and

$$\sec\theta = \frac{\sqrt{4+x^2}}{2}.$$

Example

$$\int \frac{x^2 dx}{\sqrt{9-x^2}} = \int \frac{(9\sin^2\theta)(3\cos\theta\, d\theta)}{\sqrt{9-9\sin^2\theta}}$$

$x = 3\sin\theta$

$dx = 3\cos\theta\, d\theta$

$\theta = \sin^{-1}\left(\frac{x}{3}\right)$

$\sin\theta = \frac{x}{3}$

$\cos\theta = \sqrt{1-\sin^2\theta}$

$= \sqrt{1-\frac{x^2}{9}}$

$= \frac{\sqrt{9-x^2}}{3}$

$$= \int \frac{27\sin^2\theta\cos\theta}{3\cos\theta}\, d\theta$$

$$= 9\int \sin^2\theta\, d\theta$$

$$= 9\int \frac{1-\cos 2\theta}{2}\, d\theta$$

$$= \frac{9}{2}\theta - \frac{9}{2}\frac{\sin 2\theta}{2} + C$$

$$= \frac{9}{2}\sin^{-1}\frac{x}{3} - \frac{9}{4}2\sin\theta\cos\theta + C$$

$$= \frac{9}{2}\sin^{-1}\frac{x}{3} - \frac{9}{2}\cdot\frac{x}{3}\cdot\frac{\sqrt{9-x^2}}{3} + C$$

$$= \frac{9}{2}\sin^{-1}\frac{x}{3} - \frac{x\sqrt{9-x^2}}{2} + C$$

Example

$$\int \frac{dx}{\sqrt{25x^2-4}} = \int \frac{\frac{2}{5}\sec\theta\tan\theta\, d\theta}{\sqrt{4(\sec^2\theta-1)}} = \int \frac{\frac{2}{5}\sec\theta\tan\theta\, d\theta}{2\tan\theta}$$

$$= \frac{1}{5}\int \sec\theta\, d\theta.$$

$x = \frac{2}{5}\sec\theta$

$dx = \frac{2}{5}\sec\theta\tan\theta\, d\theta$

$\sqrt{25x^2-4} = \sqrt{25\left(\frac{4}{25}\sec^2\theta\right)-4}$

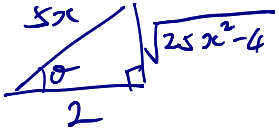
$= \sqrt{4(\sec^2\theta-1)}$

$$= \frac{1}{5}\ln|\sec\theta+\tan\theta| + C$$

$$= \frac{1}{5}\ln\left|\frac{5x}{2}+\frac{\sqrt{25x^2-4}}{2}\right| + C$$

$$= \frac{1}{5}\ln\left|5x+\sqrt{25x^2-4}\right| + C$$

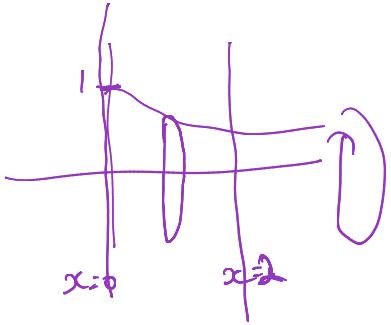
$\sec\theta = \frac{5x}{2}$



$\Rightarrow \tan\theta = \frac{\sqrt{25x^2-4}}{2}$

Example

Find the volume of the solid generated by revolving about the x-axis the region bounded by the curve $y = 4/(x^2 + 4)$, the x-axis, and the lines $x = 0$ and $x = 2$.



$$V = \int_0^2 \pi [R(x)]^2 \, dx$$

$$= \int_0^2 \pi y^2 \, dx$$

$$= \int_0^2 \pi \left(\frac{4}{x^2+4}\right)^2 dx$$

$x = 2\tan\theta$

$dx = 2\sec^2\theta \, d\theta$

when $x = 0$, $\theta = 0$

when $x = 2$, $\tan\theta = 1$

$\Rightarrow \theta = \frac{\pi}{4}$

$$= \int_0^{\frac{\pi}{4}} \pi \left(\frac{4}{4\tan^2\theta + 4}\right)^2 2\sec^2\theta \, d\theta$$

$$= \int_0^{\frac{\pi}{4}} \pi \left(\frac{1}{\sec^2\theta}\right)^2 2\sec^2\theta \, d\theta$$

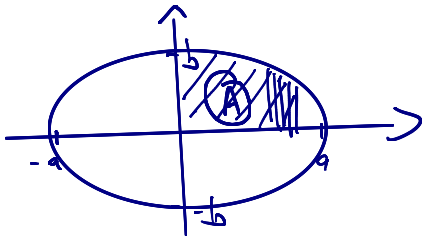
$$= 2\pi \int_0^{\frac{\pi}{4}} \cos^2\theta \, d\theta$$

$$= 2\pi \int_0^{\frac{\pi}{4}} \frac{1+\cos 2\theta}{2} \, d\theta$$

$$= \pi \left[\theta + \frac{\sin 2\theta}{2}\right]_0^{\frac{\pi}{4}} = \pi\left(\frac{\pi}{4} + \frac{1}{2}\right)$$

Example

Find the area enclosed by the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$. $(a, b > 0)$



Area of ellipse $= 4A$.

$y = b\sqrt{1 - \frac{x^2}{a^2}}$

$= \frac{b}{a}\sqrt{a^2 - x^2}$

$A = \int_0^a y\,dx = \int_0^a \frac{b}{a}\sqrt{a^2 - x^2}\,dx$

$x = a\sin\theta$

$dx = a\cos\theta\,d\theta$

when $x = 0$, $\theta = 0$

$x = a$, $\theta = \frac{\pi}{2}$

$= \int_0^{\frac{\pi}{2}} \frac{b}{a}\sqrt{a^2 - a^2\sin^2\theta}\; a\cos\theta\,d\theta$

$= b\int_0^{\frac{\pi}{2}} a\cos^2\theta\,d\theta$

$= \frac{ab}{2}\left[\theta + \frac{\sin 2\theta}{2}\right]_0^{\frac{\pi}{2}} = \frac{ab\pi}{4}$

$\Rightarrow$ Total Area $= 4A = ab\pi$

Extra Examples:

1. $\int \sqrt{9 - 4x^2}\, dx = \frac{1}{2}x\sqrt{9 - 4x^2} + \frac{9}{4}\sin^{-1}\frac{2x}{3} + C$

2. $\int_0^{\sqrt{3}/2} \frac{4x^2}{(1-x^2)^{3/2}}\, dx = 4\sqrt{3} - \frac{4\pi}{3}$

3. $\int \frac{2}{\sqrt{t}+4t\sqrt{t}}\, dt = 2\tan^{-1}(2\sqrt{t}) + C$